



**SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL
AND TECHNICAL SCIENCES**



CPU Performance Analysis

CSA1220 – COMPUTER ARCHITECTURE

Guided by:
Dr. S Radhika

Presented by:
Kolar Umme Sadiya(192425117)



INTRODUCTION



The **Central Processing Unit (CPU)** is the brain of a computer. Every task, from opening an application to running AI models, depends on the CPU's ability to execute instructions efficiently.

CPU performance is measured using:

- ❑ Clock Rate
- ❑ Clock Cycles
- ❑ Cycles Per Instruction (CPI)
- ❑ CPU Execution Time
- ❑ Benchmarking

Visual: CPU image with arrows to RAM, Storage, and Input/Output devices.



Why is CPU Performance Important?

Modern applications require high processing speed.

Challenges:

- ☐ Slow application response
- ☐ High execution time
- ☐ Increased power consumption
- ☐ Poor multitasking performance

Objective: Analyze CPU performance metrics to improve system efficiency.

Visual: Slow computer vs Fast computer comparison.



Engineer to Excel



How Does a CPU Execute Instructions?

The CPU processes every instruction in five stages:

Fetch → Decode → Execute → Memory Access → Write Back

Explanation:

- **Fetch:** Retrieves the instruction from memory.
- **Decode:** Determines the operation to perform.
- **Execute:** ALU performs the required computation.
- **Memory Access:** Reads or writes data if needed.
- **Write Back:** Stores the result in a register.
- **Visual:** Flowchart with five connected boxes.



Key CPU Performance Metrics

Clock Rate

Number of cycles executed per second.

Measured in GHz.

Clock Cycles

Number of cycles required to complete an instruction.

CPI (Cycles Per Instruction)

Average clock cycles needed for one instruction.

Formula:

$$\text{CPI} = \text{Total Clock Cycles} \div \text{Total Instructions}$$

Visual: Three connected boxes labeled Clock Rate, Clock Cycles, and CPI.

CPU Execution Time

The total time required to execute a program.

Formula

$$\text{CPU Time} = \text{Instruction Count} \times \text{CPI} \times \text{Clock Cycle Time}$$

Example

Instruction Count = 1,000,000

CPI = 2

Clock Cycle Time = 0.5 ns

CPU Time = 1 ms

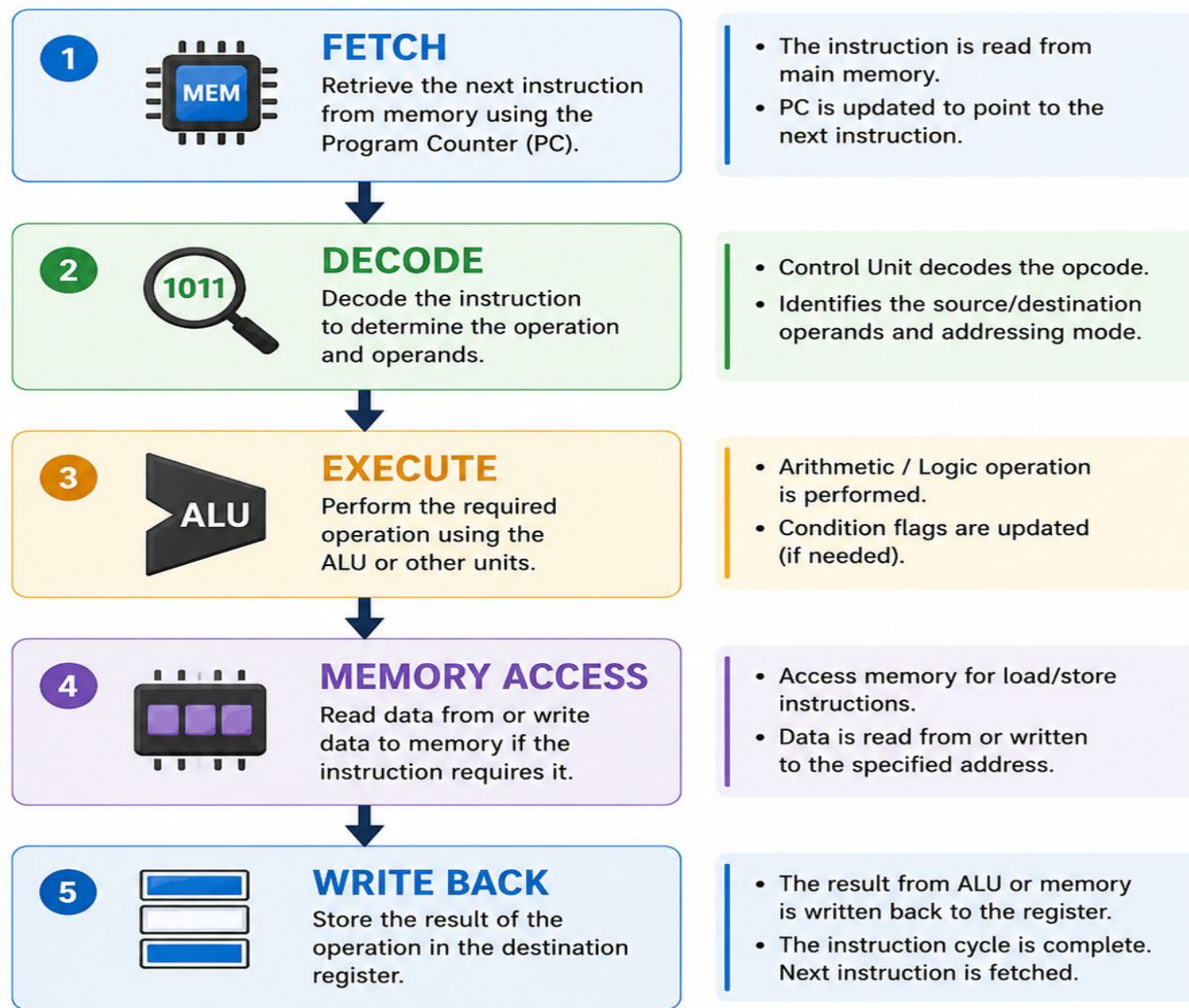
Key Point: Lower execution time means better CPU performance



Engineer to Excel



INSTRUCTION EXECUTION CYCLE



SUMMARY

The CPU repeats this cycle for each instruction in a program. This continuous cycle is the basis of all program execution.



Engineer to Excel



Measuring CPU Performance

Benchmarking compares processors using standard tests.

Popular Benchmark Tools

- ❑ SPEC CPU
- ❑ Geekbench
- ❑ Cinebench
- ❑ PassMark

Benefits

- ❑ Compares different processors
- ❑ Measures real-world performance
- ❑ Helps users choose the right CPU

Visual: Bar chart comparing benchmark scores of different CPUs.



Engineer to Excel



Where is CPU Performance Important?

- ☐ Artificial Intelligence
- ☐ Video Editing
- ☐ Gaming
- ☐ Cloud Computing
- ☐ Data Analytics
- ☐ Scientific Research
- ☐ Mobile Devices
- ☐ Autonomous Vehicles

Visual: Icons representing each application.



Engineer to Excel



How to Improve CPU Performance

- ❑ Increase Clock Speed
- ❑ Reduce CPI
- ❑ Improve Cache Memory
- ❑ Pipeline Processing
- ❑ Branch Prediction
- ❑ Out-of-Order Execution
- ❑ Hyper-Threading
- ❑ Use Faster Storage (SSD/NVMe)

Visual: Mind map connecting the techniques.



Engineer to Excel

CONCLUSION



- ❑ CPU performance is determined by multiple factors, not just clock speed.
- ❑ Lower CPI and lower execution time improve efficiency.
- ❑ Benchmarking helps evaluate and compare processors.
- ❑ Performance optimization is essential for modern computing applications.

Takeaway: Understanding CPU performance helps engineers design faster and more efficient computer systems.

THANK YOU